ALICE L. WALTON SCHOOL OF MEDICINE

Excretory Functions of the GI System

Last updated July 24, 2026 · 17 min read · CC BY-SA 4.0

Author: Ryan Downey
Editors: Jordan Finn, Russell Johnson
Contributors: Ian Murray, Eleanor Schrama, Kevin Kunkler

Listen to this Brick
0:00 / 20:27 1x

Learning Objectives (5)

After completing this brick, you will be able to:

1. Distinguish the structural organization of the large intestine and how colonic smooth muscle arrangement supports its function.
2. Explain how the large intestine absorbs water and electrolytes to form feces from liquid chyme.
3. Explain the l reflex in regulating colonic filling.
4. Trace the sequence of mass movements, the gastrocolic reflex, and the defecation reflex from stimulus to excretion.
5. Distinguish the internal from the external anal sphincter in terms of composition, innervation, and voluntary vs. involuntary control.

CASE CONNECTION

RM, a 48-year-old male, presents at the gastroenterology clinic with complaints of persistent constipation and occasional abdominal pain over the past three months. He reports straining with bowel movements and passing hard, infrequent stools. Physical examination reveals decreased bowel sounds and a palpable mass in the left lower abdomen.

What structures and mechanisms might explain RM's symptoms? How could disturbances in colonic motility contribute to constipation and a palpable abdominal mass? Consider your answers as you read, and we'll revisit RM at the end of the brick.

GO TO CONCLUSION ↓

Structural Organization of the Large Intestine (LO1)

The large intestine differs from the small intestine in both structure and function. While the small intestine is specialized for nutrient digestion and absorption via villi a imized for water and electrolyte absorption, storage, and controlled expulsion of waste. These functional differences are reflected in its unique histological and muscular architecture.

Muscle Layers and Taeniae Coli

Like the rest of the gastrointestinal (GI) tract, the large intestine contains a muscularis externa composed of an inner circular muscle layer and an outer longitudinal muscle layer. However, a defining feature of the colon is that the outer longitudinal layer is not continuous. Instead, it condenses into three distinct bands called the taeniae coli (TC), which run along the length of the colon. These bands are shorter than the colon itself, creating a puckering effect along the colonic wall.

The TC play a critical role in colonic motility. Their tonic contraction generates tension along the colon, contributing to the segmental contractions that mix and slowly propel luminal contents. This arrangement allows the colon to perform its primary motility functions—segmentation and mass movement—without requiring the continuous longitudinal muscle sheet seen in the small intestine.

Haustra Formation and Function

The puckering caused by the TC produces outward bulges in the colonic wall known as haustra (singular: haustrum). Haustra are visible both radiographically and endoscopically and represent a hallmark of normal colonic anatomy. They are not permanent structures—they shift as the circular muscle contracts and relaxes in different segments.

Functionally, haustra mixing and temporary storage of fecal matter. Haustral contractions churn luminal contents, maximizing contact with the absorptive epithelium and facilitating water and electrolyte extraction. This slow, mixing motility pattern is well-suited to the colon's role in gradually converting liquid chyme into formed stool over a transit time of 24–72 hours.

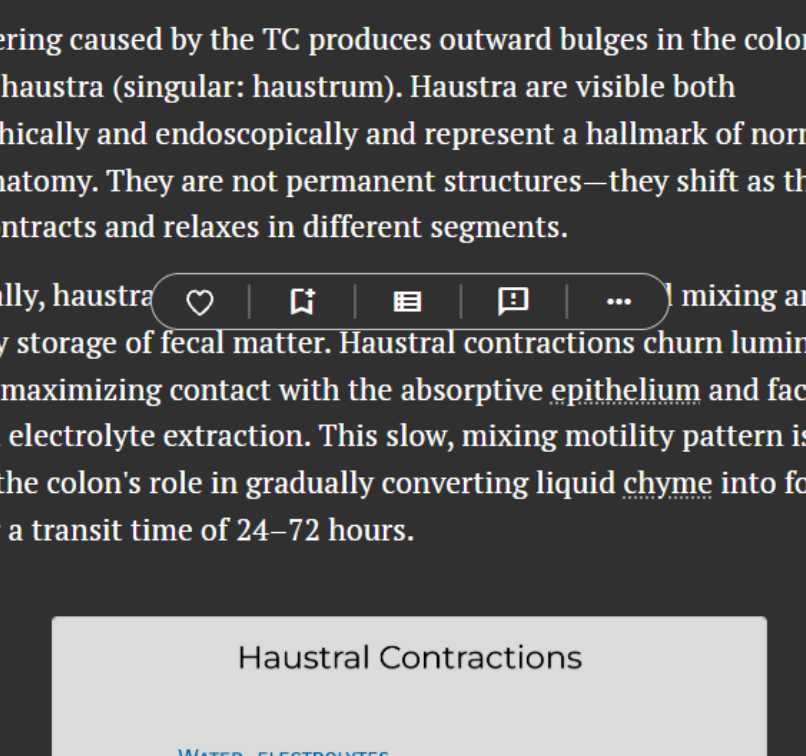

Figure 1: The taeniae coli gather the colonic wall into sacculations called haustra. Haustral contractions of the circular muscle churn contents within these compartments, maximizing mucosal contact for water and electrolyte absorption while slowly mixing and propelling fecal matter.

What structural features unique to the colon are formed by the condensation of the outer longitudinal muscle layer, and what do they produce?

Colonic Water and Electrolyte Absorption (LO2)

Approximately 1–2 liters of liquid chyme enter the colon daily from the ileum. By the time stool is excreted, only about 100–200 mL of water remains. This dramatic reduction is achieved through highly efficient epithelial transport mechanisms distributed along the colonic mucosa. Understanding these mechanisms is essential for explaining common clinical conditions such as diarrhea and constipation.

Epithelial Transport Mechanisms

Sodium (Na^+) absorption is the primary driver of water reabsorption in the colon. Na^+ enters colonocytes via apical sodium channels (ENaC) and Na^+/H^+ exchangers (NHE), then is actively pumped out basolaterally by Na^+/K^+-ATPase. This creates an osmotic gradient that draws water paracellularly into the submucosa. Aldoster easing Na^+ and water absorption—a key hormonal control point.

Chloride (Cl^-) absorption occurs via apical Cl^-/HCO_3^- exchangers, which simultaneously secrete bicarbonate into the lumen—explaining the alkaline nature of colonic secretions. Short-chain fatty acids (SCFAs), produced by bacterial fermentation of dietary fiber, also stimulate Na^+ and water absorption, linking the gut microbiome to colonic fluid balance. Potassium (K^+) is both absorbed and secreted by the colon depending on luminal concentrations and hormonal status.

Formation of Feces

As chyme progresses from the cecum through the ascending, transverse, and descending colon, progressive water and electrolyte absorption transforms it from a liquid to a semisolid consistency. By the time contents reach the sigmoid colon and rectum, formed stool consists of approximately 75% water, with the remainder composed of bacteria, undigested fiber, desquamated epithelial cells, and bile pigments.

CLINICAL CORRELATION

Disruption of these absorption mechanisms has direct clinical consequences. In secretory diarrhea, excessive Cl^- secretion (e.g., via cAMP-activated CFTR channels in cholera) overwhelms absorptive capacity, producing watery stools. Conversely, prolonged colonic transit—as in slow-transit constipation—allows excessive water extraction, resulting in hard, dry stools that are difficult to pass. This is clinically relevant when evaluating patients like RM.

ACTIVE LEARNING QUESTION

A patient with chronic diarrhea has low stool osmotic gap. Which mechanism best explains increased stool water content in secretory diarrhea?

A. ○ Increased ENaC activity
B. ○ Increased Cl^- secretion
C. ○ Aldosterone excess
D. ○ Decreased NHE activity
E. ○ Increased SCFAs

Explanation (requires correct answer)

The Ileocecal Valve and Gastroileal Reflex (LO3)

The transition from the small intestine to the large intestine is not simply anatomical—it is a regulated gateway. The ileocecal valve (ICV) and the gastroileal reflex work in concert to control the rate and timing of chyme delivery into the colo that the colon is not overwhelmed with luminal contents.

Valve Structure and Function

The ICV is a sphincter-like structure located at the junction of the terminal ileum and cecum. It consists of two lips of mucosa reinforced by thickened circular smooth muscle that maintains a state of tonic contraction at rest, keeping the valve closed. This prevents retrograde flow of colonic bacteria-laden contents back into the ileum—a critical defense against bacterial overgrowth in the small intestine.

The ICV opens in response to peristaltic waves arriving from the ileum and to distension of the ileum itself. Increased pressure in the ileum relaxes the valve, allowing small boluses of chyme to pass into the cecum. Conversely, distension of the cecum triggers contraction of the ICV, slowing further entry—a feedback mechanism that prevents colonic overload.

CLINICAL CORRELATION

Clinically, resection of the ICV (e.g., in right hemicolectomy) can lead to diarrhea and bacterial overgrowth.

Gastroileal Reflex Pathways

The gastroileal reflex is a neurohormonal response that coordinates ileal emptying with gastric activity. When food enters the stomach and causes gastric distension, si ervous system (ENS) and extrinsic autonomic pathways to increase peristalsis in the ileum. This accelerates the movement of existing ileal contents toward the ICV, promoting colonic filling in anticipation of the incoming meal.

Gastrin, released by G cells in the gastric antrum in response to a meal, also contributes to this reflex by enhancing ileal motility. The net effect is a coordinated "make room" response: as the stomach fills, the ileum empties into the colon. This reflex explains why many individuals feel the urge to defecate shortly after eating—a phenomenon that also involves the gastrocolic reflex discussed in the next section.

ACTIVE LEARNING QUESTION

A patient undergoes surgical resection of the ileocecal valve. Which complication is most directly attributable to loss of ICV function?

A. ○ Increased gastric acid secretion
B. ○ Reduced bile acid reabsorption
C. ○ Small intestinal bacterial overgrowth
D. ○ Impaired colonic water absorption
E. ○ Decreased gastrin secretion

Explanation

Mass Movements, the Gastrocolic Reflex, and Defecation (LO4)

Colonic motility is not a single, continuous process—it is a sequence of coordinated events triggered by specific stimuli and mediated by both intrinsic and extrinsic neural pathways. From the powerful propulsive contractions that move stool toward the rectum, to the complex reflex arc that culminates in defecation, each step is precisely regulated to ensure efficient and socially appropriate waste elimination.

Mass Movement Initiation

Mass movements (MM) are powerful, propulsive contractions that sweep colonic contents over long distances—sometimes spanning one-third to one-half of the colon in a single contraction. Unlike haustral contractions, which are localized and mixing in nature, MM are characterized by simultaneous contraction of a long colonic segment with relaxation ahead of it, rapidly advancing fecal material toward the rectum.

MM occur infrequently—typically 1–3 times per day—and are most commonly triggered by gastric or duodenal distension following a meal. They are mediated by the ENS and modulated by extrinsic autonomic input. The high-amplitude contractions of MM are responsible for the sensation of urgency that often precedes defecation. Reduced frequency or amplitude of MM is a key mechanism in slow-t

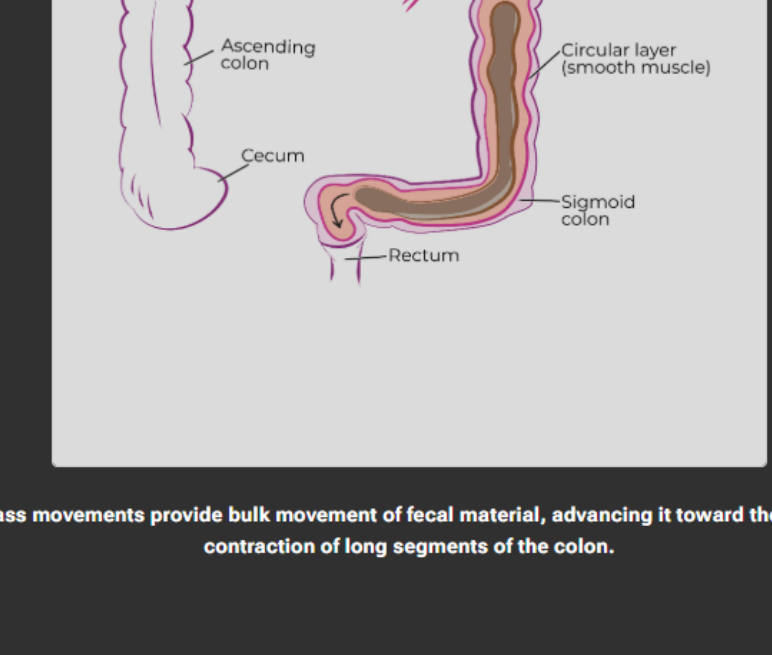

Figure 2: Mass movements provide bulk movement of fecal material, advancing it toward the rectum through contraction of long segments of the colon.

Gastrocolic and Defecation Reflexes

The gastrocolic reflex (GCR) is initiated when food enters the stomach, causing gastric distension. This triggers both ENS-mediated and extrinsic neural signals (via the vagus nerve and pelvic nerves) that dramatically increase colonic motility, including the initiation of MM. Hormones such as gastrin and cholecystokinin (CCK) amplify this response. The GCR explains the common postprandial urge to onditions like irritable bowel syndrome (IBS).

When MM propel fecal material into the rectum, rectal distension triggers the defecation reflex. This reflex involves two components: an intrinsic reflex mediated by the ENS (myenteric plexus) that causes relaxation of the internal anal sphincter (IAS) via the rectoanal inhibitory reflex (RAIR), and an extrinsic reflex mediated by sacral parasympathetic nerves (S2–S4) that augments rectal contraction. The sequence of events in the defecation reflex proceeds as follows:

Steps 1–3: intrinsic and extrinsic reflex components

1. Fecal matter enters the rectum, causing rectal wall distension
2. Stretch receptors activate the ENS, triggering the RAIR and IAS relaxation
3. Sacral parasympathetic signals increase rectal peristalsis

Steps 4–5: voluntary augmentation

4. Voluntary relaxation of the external anal sphincter (EAS) allows defecation
5. Valsalva maneuver (increased intra-abdominal pressure) facilitates expulsion

What is the sequence of the defecation reflex from rectal distension to stool expulsion?

Internal vs. External Anal Sphincters (LO5)

Continence and defecation depend on the coordinated action of two anatomically and functionally distinct sphincters at the anal canal. The internal anal sphincter (IAS) and external anal sphincter (EAS) differ in muscle type, innervation, and the degree to which they are under voluntary control. Understanding these differences is essential for diagnosing and managing disorders of continence and defecation.

Muscle Composition

The IAS is composed of smooth muscle and represents a thickened continuation of the inner circular muscle layer of the rectum. As smooth muscle, it operates involuntarily and maintains a state of tonic contraction at rest, providing approximately 70–85% of resting anal canal pressure. This sustained tone is critical for passive continence—preventing leakage without conscious effort.

The EAS, in contrast, is composed of skeletal (striated) muscle and forms a ring around the IAS. As skeletal muscle, it is capable of rapid, forceful voluntary contraction. The EAS contributes to continence during moments of increased intra-abdominal pres n be voluntarily contracted to defer defecation when socially inconvenient. Its resting tone is lower than the IAS but increases significantly with voluntary effort.

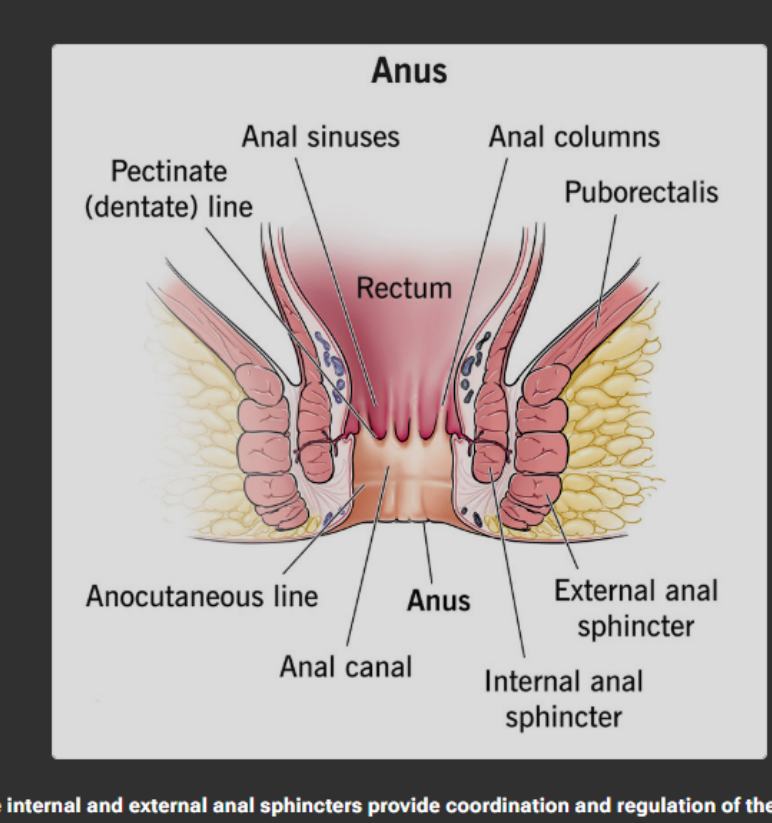

Figure 3: The internal and external anal sphincters provide coordination and regulation of the final steps of the fecal elimination pathway.

Innervation and Control

The IAS receives its innervation from the autonomic nervous system. Sympathetic fibers (L1–L2, via the hypogastric nerve) promote IAS contraction, maintaining continen a the pelvic nerve) cause IAS relaxation, facilitating defecation. The rectoanal inhibitory reflex (RAIR)—relaxation of the IAS in response to rectal distension—is an intrinsic ENS-mediated reflex that occurs even in the absence of extrinsic innervation. Absence of the RAIR is the hallmark of Hirschsprung disease.

CLINICAL CORRELATION

Hirschsprung Disease

Hirschsprung disease results from the congenital absence of enteric ganglion cells (aganglionosis) in a segment of the distal colon, most commonly the rectosigmoid. Without ENS innervation, the affected segment cannot relax, creating a functional obstruction. The hallmark diagnostic finding is an **absent rectoanal inhibitory reflex (RAIR)** on anorectal manometry — the IAS fails to relax in response to rectal distension because the intrinsic reflex arc is missing. Proximal to the aganglionic segment, the normally innervated colon dilates (megacolon) as stool accumulates.

The EAS is innervated by the pudendal nerve (S2–S4), a somatic nerve that enables voluntary control. Cortical input via the corticospinal tract allows conscious contraction of the EAS to maintain continence. Damage to the pudendal nerve—as can occur in childbirth or pelvic surgery—results in fecal incontinence due to loss of voluntary sphincter control. The table below summarizes the key differences between the two sphincters:

Feature	Internal Anal Sphincter (IAS)	External Anal Sphincter (EAS)
Muscle Type	Smooth muscle	Skeletal (striated) muscle
Control	Involuntary	Voluntary
Innervation	Autonomic (sympathetic L1–L2; parasympathetic S2–S4) + ENS	Somatic (pudendal nerve, S2–S4)
Resting Tone	High (70–85% of resting pressure)	Lower; increases with voluntary effort
Key Reflex	Rectoanal inhibitory reflex (RAIR)	Voluntary contraction to defer defecation

CLINICAL CORRELATION

Fecal Incontinence — IAS vs. EAS Failure

Fecal incontinence results from failure of one or both anal sphincters, but the clinical pattern differs depending on which sphincter is affected. **IAS dysfunction** (e.g., from smooth muscle degeneration or autonomic neuropathy) causes **passive incontinence** — involuntary leakage at rest, particularly of liquid stool or gas, because resting anal canal pressure is lost. **EAS dysfunction** (e.g., from pudendal nerve injury during childbirth or pelvic surgery) causes **urge incontinence** — the patient senses rectal filling but cannot voluntarily defer defecation. Anorectal manometry distinguishes the two by measuring resting pressure (IAS) and squeeze pressure (EAS).

ACTIVE LEARNING QUESTION

A patient with Hirschsprung disease lacks enteric ganglion cells in the rectosigmoid. Which finding is the direct consequence of this defect?

A. ○ Absent rectoanal inhibitory reflex
B. ○ Loss of voluntary EAS contraction
C. ○ Increased colonic secretion
D. ○ Loss of gastrocolic reflex
E. ○ Impaired haustral contractions

Explanation (requires correct answer)

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to RM, what structures and motility functions could explain his constipation and abdominal findings? How would disruption in colonic reflexes lead to these symptoms?

RM's persistent constipation and hard stools likely reflect reduced frequency or amplitude of mass movements, leading to prolonged colonic transit and excessive water absorption fro dant mass is consistent with a fec on where formed stool accumulates when propulsive motility is impaired. Decreased bowel sounds suggest reduced overall colonic motility, possibly due to dysfunction in the gastrocolic reflex or ENS signaling. If the defecation reflex is also impaired—for example, due to blunted rectal sensation or IAS dysfunction—stool would accumulate further, worsening obstruction. Evaluation of RM should include assessment of colonic transit time and anorectal manometry to test the RAIR.

Summary

- The large intestine is structurally distinct from the small intestine: its outer longitudinal muscle condenses into three taeniae coli (TC), which pucker the wall into haustra that facilitate segmental mixing and slow fecal transit.
- The colon absorbs 1–2 L of water daily via Na^+-driven osmotic gradients (ENaC, NHE) and Cl^-/HCO_3^- exchange, converting liquid chyme into formed stool; disruption causes diarrhea or constipation.
- The ileocecal valve (ICV) maintains a tonic closure that prevents bacterial backflow into the ileum and regulates chyme entry into the cecum based on ileal and cecal pressure.
- The gastroileal reflex, triggered by gastric distension and amplified by gastrin, accelerates ileal peristalsis to promote colonic filling after a meal.
- Mass movements (1–3/day) propel colonic contents toward the rectum; the gastrocolic reflex (mediated by ENS, vagus, and hormones like CCK) triggers these movements postprandially.
- The defecation reflex involves rectal distension → ENS-mediated IAS relaxation (RAIR tion → voluntary EAS relaxation → expulsion; absence of the RAIR is the hallmark of Hirschsprung disease.
- The IAS (smooth muscle, autonomic/ENS innervation, involuntary) provides resting continence, while the EAS (skeletal muscle, pudendal nerve, voluntary) allows conscious deferral of defecation; damage to either causes incontinence.

Review Questions

1. What is the role of taeniae coli in the colon?

A. ○ Inner circular muscle layer
B. ○ Continuous outer longitudinal muscle
C. ○ Longitudinal bands forming haustra
D. ○ Permanent bulges in colon wall

Explanation (requires correct answer)

2. Which mechanism primarily drives water absorption in the colon?

A. ○ Sodium absorption via ENaC and NHE
B. ○ Chloride secretion into lumen
C. ○ Aldosterone excess
D. ○ Decreased NHE activity
E. ○ Short-chain fatty acid stimulation

Explanation (requires correct answer)

3. What is a direct effect of ileocecal valve loss after surgery?

A. ○ Increased gastric acid secretion
B. ○ Reduced bile acid reabsorption
C. ○ Small intestinal bacterial overgrowth
D. ○ Impaired colonic water absorption
E. ○ Decreased gastrin secretion

Explanation (requires correct answer)

4. Which sequence correctly describes the defecation reflex?

A. ○ Voluntary EAS relaxation first
B. ○ Rectal dis
C. ○ Valsalva precedes IAS relaxation
D. ○ Sacral nerves relax EAS
E. ○ ENS inhibits sacral pathways

Explanation (requires correct answer)

5. Which distinguishes the internal anal sphincter from the external?

A. ○ Skeletal muscle with somatic control
B. ○ Voluntary contraction and low resting tone
C. ○ Innervated by pudendal nerve
D. ○ Smooth muscle with involuntary control
E. ○ Lower resting tone than EAS

Explanation (requires correct answer)

How would you rate this Brick?

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ Distinguish the structural organization of the large intestine and how colonic smooth muscle arrangement supports its function.
02 ☐ Explain how the large intestine absorbs water and electrolytes to form feces from liquid chyme.
03 ☐ Explain the role of the ileocecal valve and gastroileal reflex in regulating colonic filling.
04 ☐ Trace the sequence of mass movements, the gastrocolic reflex, and the defecation reflex from stimulus to excretion.
05 ☐ Distinguish the internal from the external anal sphincter in terms of composition, innervation, and voluntary vs. involuntary control.

Objective Confidence: 0/5